

```
In [1]: import pandas as pd

df= pd.read_csv("customer_shopping_behavior.csv")
```

```
In [2]: df.head()
```

Out[2]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Active
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Active
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Active
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Active
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Active

```
In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                      3900 non-null   object
4   Category                            3900 non-null   object
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   object
7   Size                                3900 non-null   object
8   Color                               3900 non-null   object
9   Season                              3900 non-null   object
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   object
12  Shipping Type                      3900 non-null   object
13  Discount Applied                   3900 non-null   object
14  Promo Code Used                    3900 non-null   object
15  Previous Purchases                 3900 non-null   int64
16  Payment Method                     3900 non-null   object
17  Frequency of Purchases              3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
In [4]: df.describe(include='all')
```

Out[4]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5

```
In [5]: df.isnull().sum()
```

```
Out[5]: Customer ID      0
      Age                0
      Gender             0
      Item Purchased     0
      Category           0
      Purchase Amount (USD) 0
      Location           0
      Size               0
      Color              0
      Season             0
      Review Rating      37
      Subscription Status 0
      Shipping Type      0
      Discount Applied   0
      Promo Code Used    0
      Previous Purchases 0
      Payment Method     0
      Frequency of Purchases 0
      dtype: int64
```

```
In [6]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
In [7]: df.isnull().sum()
```

```
Out[7]: Customer ID      0
      Age                0
      Gender             0
      Item Purchased     0
      Category           0
      Purchase Amount (USD) 0
      Location           0
      Size               0
      Color              0
      Season             0
      Review Rating      0
      Subscription Status 0
      Shipping Type      0
      Discount Applied   0
      Promo Code Used    0
      Previous Purchases 0
      Payment Method     0
      Frequency of Purchases 0
      dtype: int64
```

```
In [8]: df.columns = df.columns.str.lower()
      df.columns = df.columns.str.replace(' ', '_')
      df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
In [9]: df.columns
```

```
Out[9]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

```
In [10]: # create a column age_group
      labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
      df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

```
In [11]: df[['age', 'age_group']].head(10)
```

Out[11]:

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

```
In [12]: # create column purchase_frequency_days

frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days']=df['frequency_of_purchases'].map(frequency_mapping)
```

```
In [13]: df[['purchase_frequency_days','frequency_of_purchases']]
```

Out[13]:

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
...	...	...
3895	7	Weekly
3896	14	Bi-Weekly
3897	90	Quarterly
3898	7	Weekly
3899	90	Quarterly

3900 rows × 2 columns

```
In [14]: df[['discount_applied','promo_code_used']].head(10)
```

```
Out[14]:
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
In [15]: (df['discount_applied'] == df['promo_code_used']).all()
```

```
Out[15]: np.True_
```

```
In [16]: df = df.drop('promo_code_used',axis=1)
```

```
In [17]: df.columns
```

```
Out[17]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
               'purchase_amount', 'location', 'size', 'color', 'season',
               'review_rating', 'subscription_status', 'shipping_type',
               'discount_applied', 'previous_purchases', 'payment_method',
               'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
              dtype='object')
```

```
In [18]: !pip install psycopg2-binary
```

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: psycopg2-binary in C:\Users\Gamer\AppData\Roaming\Python\Python314\site-packages (2.9.12)
Note: you may need to restart the kernel to use updated packages.

```
from sqlalchemy import create_engine
```

```
username = "****" password = "*****" host = "*****" port = "*****" database = "customer_behavior"
```

```
engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")
```

load dataframe into postgresql

```
table_name = "customer" df.to_sql(table_name, engine, if_exists="replace", index=False)
```

```
print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")
```

```
In [21]: import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
plt.rcParams['figure.figsize'] = (10, 5)
```

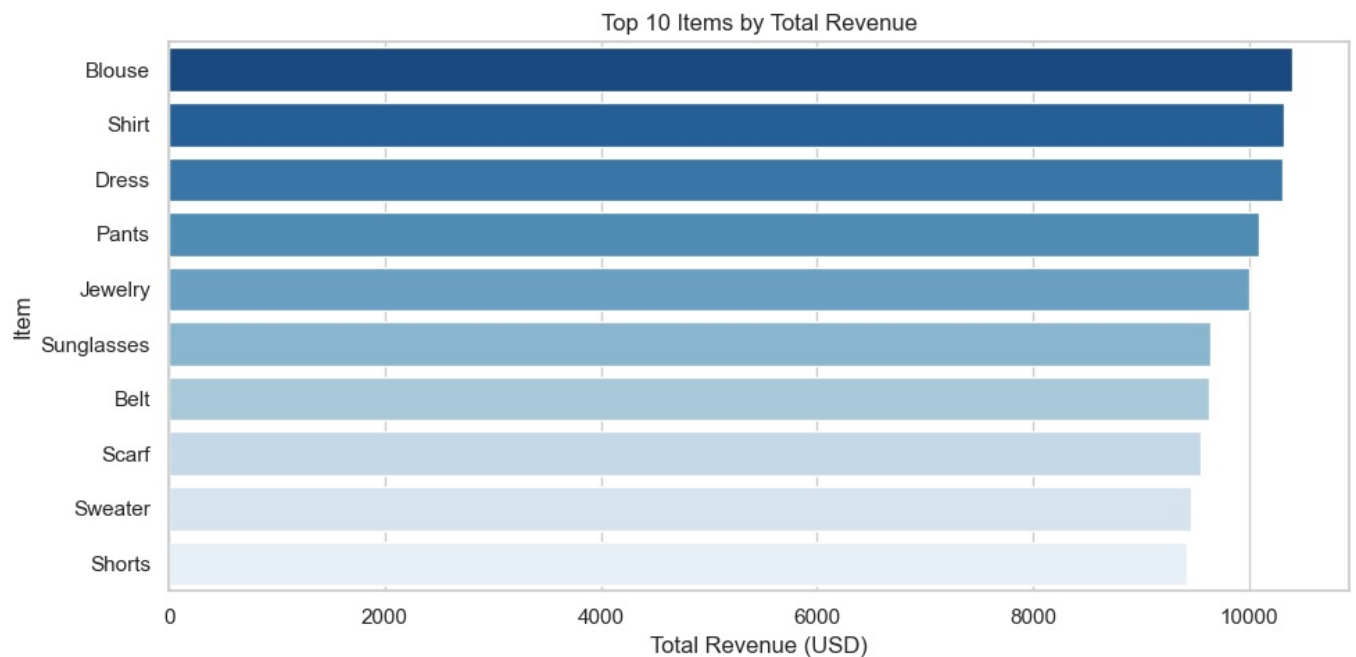
```
In [22]: # Purchase Amount Distribution
plt.figure()
sns.histplot(df['purchase_amount'], bins=30, kde=True, color='steelblue')
plt.title('Purchase Amount Distribution')
plt.xlabel('Purchase Amount (USD)')
plt.ylabel('Count')
plt.tight_layout()
plt.savefig('purchase_amount_distribution.png', dpi=150)
plt.show()
```



In [37]: # Top 10 Items by Revenue

```
top_items = df.groupby('item_purchased')['purchase_amount'].sum().sort_values(ascending=False).head(10)

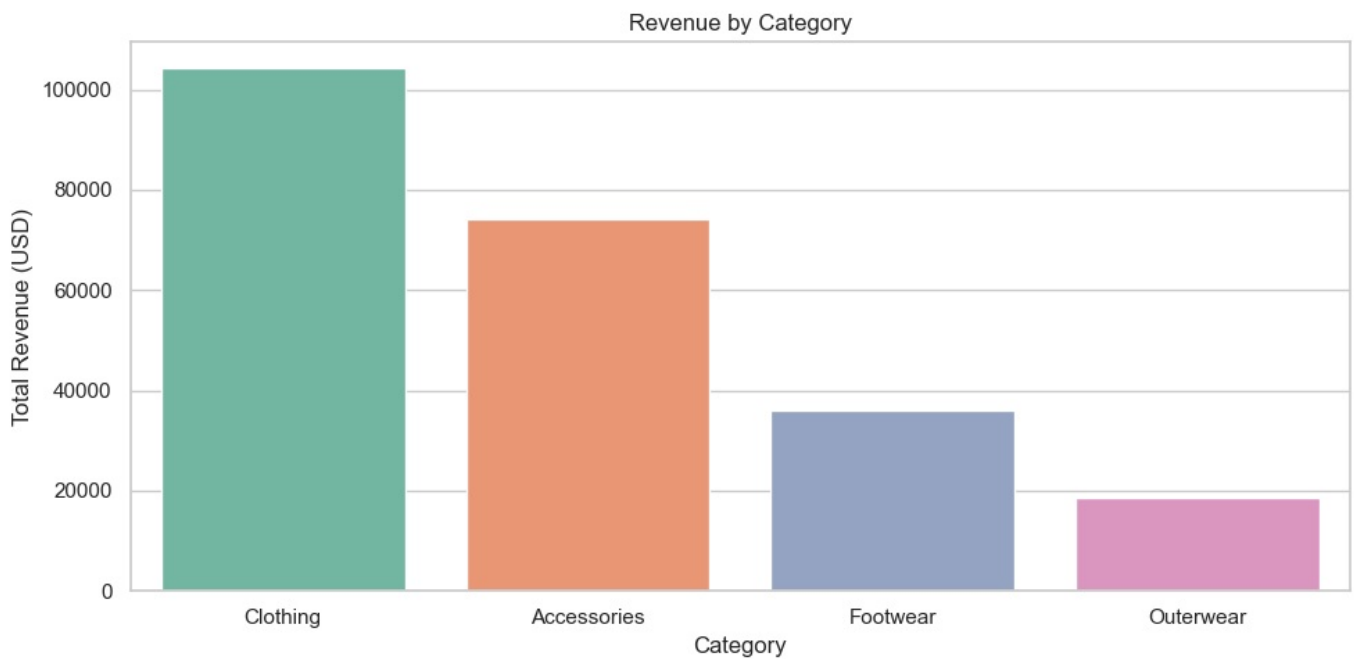
plt.figure()
sns.barplot(x=top_items.values, y=top_items.index, hue=top_items.index, palette='Blues_r')
plt.title('Top 10 Items by Total Revenue')
plt.xlabel('Total Revenue (USD)')
plt.ylabel('Item')
plt.tight_layout()
plt.savefig('top_items_revenue.png', dpi=150)
plt.show()
```



In [38]: # Category-wise Revenue

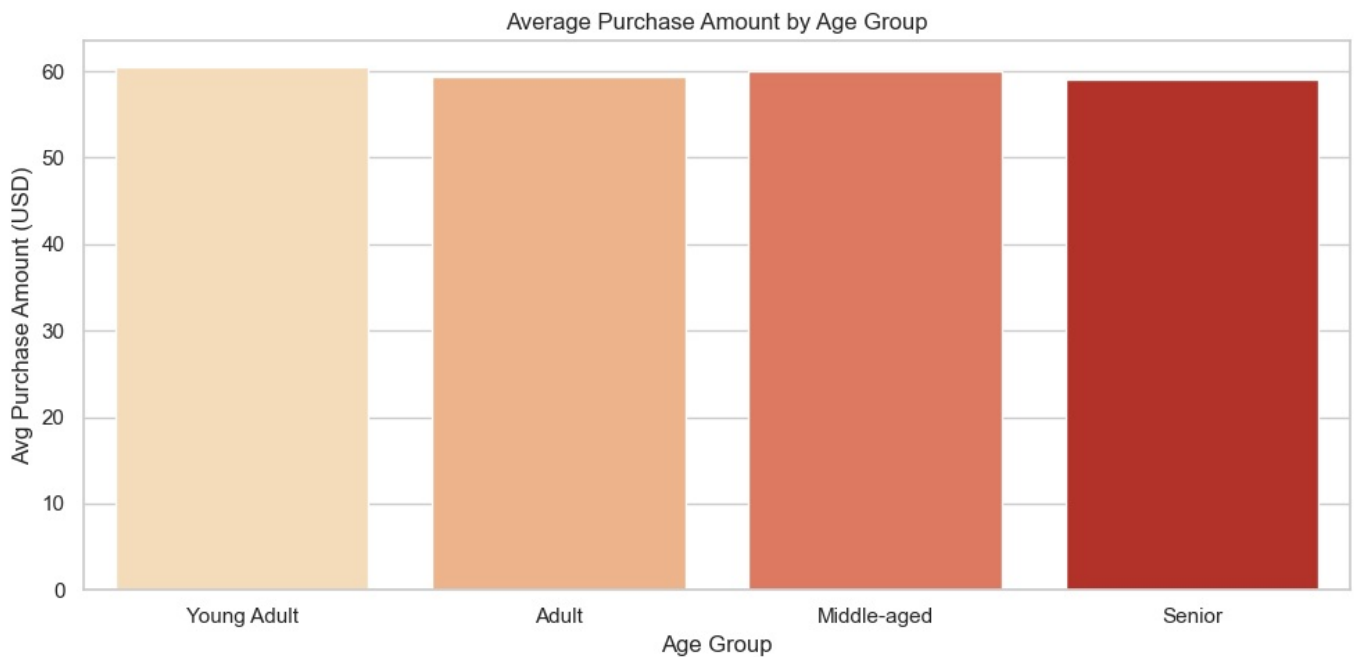
```
cat_rev = df.groupby('category')['purchase_amount'].sum().sort_values(ascending=False)

plt.figure()
sns.barplot(x=cat_rev.index, y=cat_rev.values, hue=cat_rev.index, palette='Set2')
plt.title('Revenue by Category')
plt.xlabel('Category')
plt.ylabel('Total Revenue (USD)')
plt.tight_layout()
plt.savefig('category_revenue.png', dpi=150)
plt.show()
```



```
In [40]: # Age Group vs Avg Purchase Amount
age_purchase = df.groupby('age_group', observed=True)['purchase_amount'].mean().reset_index()

plt.figure()
sns.barplot(data=age_purchase, x='age_group', y='purchase_amount', hue='age_group', palette='OrRd')
plt.title('Average Purchase Amount by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Avg Purchase Amount (USD)')
plt.tight_layout()
plt.savefig('age_group_purchase.png', dpi=150)
plt.show()
```



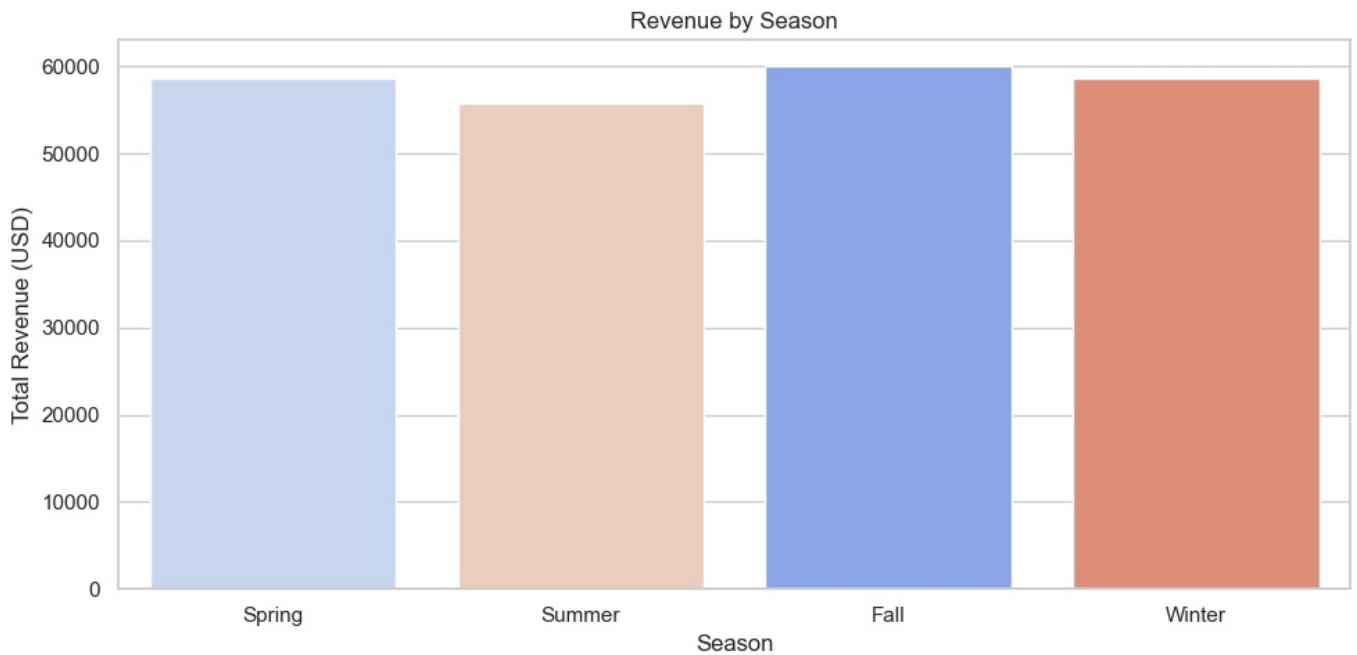
```
In [43]: # Gender vs Purchase Amount

plt.figure()
sns.boxplot(data=df, x='gender', y='purchase_amount', hue='gender', palette='pastel')
plt.title('Purchase Amount by Gender')
plt.xlabel('Gender')
plt.ylabel('Purchase Amount (USD)')
plt.tight_layout()
plt.savefig('gender_purchase.png', dpi=150)
plt.show()
```



```
In [44]: # Season-wise Revenue
season_rev = df.groupby('season')['purchase_amount'].sum().reset_index()

plt.figure()
sns.barplot(data=season_rev, x='season', y='purchase_amount', hue='season',
            order=['Spring', 'Summer', 'Fall', 'Winter'], palette='coolwarm')
plt.title('Revenue by Season')
plt.xlabel('Season')
plt.ylabel('Total Revenue (USD)')
plt.tight_layout()
plt.savefig('season_revenue.png', dpi=150)
plt.show()
```

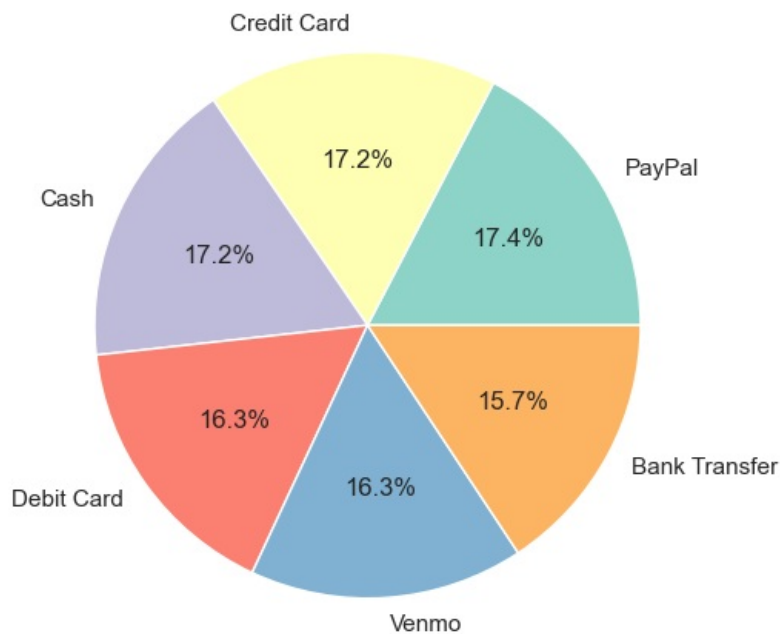


```
In [30]: # Payment Method Distribution

payment_counts = df['payment_method'].value_counts()

plt.figure()
plt.pie(payment_counts.values, labels=payment_counts.index, autopct='%1.1f%%',
        colors=sns.color_palette('Set3', len(payment_counts)))
plt.title('Payment Method Distribution')
plt.tight_layout()
plt.savefig('payment_methods.png', dpi=150)
plt.show()
```

Payment Method Distribution



In [45]: # Subscription vs Avg Purchase

```
sub_avg = df.groupby('subscription_status')['purchase_amount'].mean().reset_index()

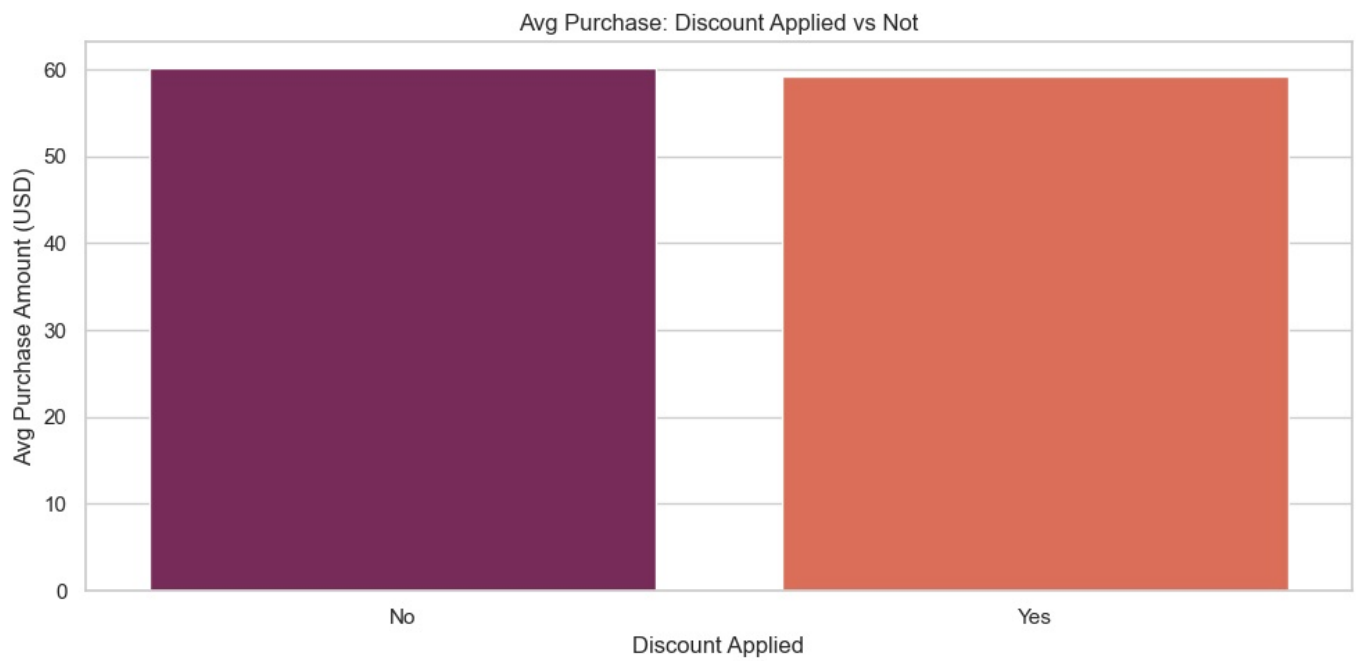
plt.figure()
sns.barplot(data=sub_avg, x='subscription_status', y='purchase_amount', hue='subscription_status', palette='muted')
plt.title('Avg Purchase Amount: Subscriber vs Non-Subscriber')
plt.xlabel('Subscription Status')
plt.ylabel('Avg Purchase (USD)')
plt.tight_layout()
plt.savefig('subscription_purchase.png', dpi=150)
plt.show()
```



In [46]: # Discount vs No Discount Revenue

```
disc_rev = df.groupby('discount_applied')['purchase_amount'].mean().reset_index()

plt.figure()
sns.barplot(data=disc_rev, x='discount_applied', y='purchase_amount', hue='discount_applied', palette='rocket')
plt.title('Avg Purchase: Discount Applied vs Not')
plt.xlabel('Discount Applied')
plt.ylabel('Avg Purchase Amount (USD)')
plt.tight_layout()
plt.savefig('discount_impact.png', dpi=150)
plt.show()
```

```
In [33]: # Correlation Heatmap (Numeric columns)
numeric_df = df.select_dtypes(include='number')

plt.figure(figsize=(8, 6))
sns.heatmap(numeric_df.corr(), annot=True, fmt='.2f', cmap='coolwarm', linewidths=0.5)
plt.title('Correlation Heatmap')
plt.tight_layout()
plt.savefig('correlation_heatmap.png', dpi=150)
plt.show()
```

